

NIKHIL SAXENA

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Education

Dr. A.P.J. Abdul Kalam Technical University

Sep. 2023 – Expected May 2027

Bachelor of Technology, Computer Engineering

Projects

High-Throughput Notification Broker | Go, gRPC, Redis, Kubernetes, Prometheus, OpenTelemetry May 2026

- Architected a distributed notification engine in Go and gRPC, sustaining **2,900+ RPS** with **<55ms P95** latency across a decoupled **3-tier microservice** architecture.
- Developed a fault-tolerant, zero-loss queue utilizing Redis BLMOVE for atomic processing, integrating priority-based exponential backoff, jitter, and automated Dead Letter Queue (DLQ) escalation to guarantee message delivery.
- Accelerated SDLC execution via **Cursor** and **Claude** AI-assisted workflows, automating the scaffolding of gRPC protocol buffers, Prometheus/Grafana dashboards, and OpenTelemetry tracing across asynchronous Redis boundaries via custom context carriers.
- Streamlined deployment and security using multi-stage Distless Docker builds (reducing image size to **<20MB**), orchestrated via Kubernetes Deployments, StatefulSets, and dynamic Horizontal Pod Autoscalers (HPA).
- Deployed to AWS (EC2, ECR, S3) using IAM instance profiles for zero-hardcoded-credential access; engineered a background DLQ archiver polling at **60s** intervals — draining **100 jobs/cycle** into date-partitioned S3 JSON archives — tracked via dedicated Prometheus counters for archive health.

Meta Clash | Go, Next.js, PostgreSQL, WebSocket, Docker, GitHub Actions, JWT | Live Demo Dec. 2025

- Designed a server-authoritative Go backend with hexagonal architecture and a **3-state FSM**, managing deterministic combat for **4 concurrent players** over a thread-safe WebSocket hub (sync.RWMutex, goroutines, channels) with a **54s** ping/pong heartbeat.
- Integrated a custom **4-agent AI workflow** (Architect, Developer, Reviewer, QA) via **Claude MCP** into the SDLC, automating code reviews, generating structured changelogs, and leveraging **Graphify** for persistent memory retention across iterative engineering cycles.
- Built a fault-tolerant generation pipeline integrating a Jikan REST API client, a thread-safe in-memory TTL cache (**100-entry cap, 1h TTL**), and **Gemini 2.5 Flash**, with automatic fallback to deterministic FNV-1a hashing on API timeouts.
- Integrated stateless JWT authentication and a PostgreSQL database via the Repository pattern, handling connection pooling and automated schema migrations for persistent match history.
- Deployed via multi-stage Docker builds to Railway and Vercel, establishing a GitHub Actions CI pipeline for container builds, go vet, and Go -race condition testing.

Technical Skills

Languages: Go, C++, Python, SQL (PostgreSQL), JavaScript/TypeScript, Bash/Shell

AI Engineering & Tooling: Claude API, Model Context Protocol (MCP), Cursor, Prompt Engineering, Multi-Agent Workflows, Graphify, Codex

Backend, Cloud & DevOps: AWS (EC2, ECR, S3), gRPC, RESTful APIs, WebSockets, Redis, Docker, Kubernetes, Next.js, JWT

Observability & Tools: Prometheus, Grafana, OpenTelemetry, Git, GitHub Actions, Postman, Linux (Pop!_OS)

CS & Paradigms: Distributed Systems, Data Structures & Algorithms, OOP, OS, DBMS, Computer Networks, Microservices, Hexagonal Architecture, Concurrency

Experience

Creative and Tourism Club

Aug. 2024 – May 2026

General Secretary

ABES Engineering College

- Led a **35+ member team** to execute college-wide events (“ABES Got Talent”, “SAARANG’26”) for **500+ attendees** across a **2-month timeline**, managing an **INR 55,000 budget** and securing **Royal Enfield** sponsorship (INR 7,000 in prizes).

Atlas Mapper | @atlasmapper

Nov. 2020 – Sep. 2023

Independent Digital Creator

- Scaled a digital geography platform to **20K+ followers** and **3M monthly interactions** through custom infographics, negotiating and managing end-to-end **paid promotional campaigns** with educational clients.